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(54) **IMAGE RECORDING APPARATUS**

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See application file for complete search history.

(71) Applicant: **BROTHER KOGYO KABUSHIKI**
KAISHA, Nagoya (JP)

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(72) Inventors: **Masatomo Yamaguchi**, Inazawa (JP);
Yuya Tatematsu, Nagoya (JP)

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(73) Assignee: **BROTHER KOGYO KABUSHIKI**
KAISHA, Nagoya (JP)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 196 days.

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Primary Examiner — Leslie A Nicholson, III

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(74) *Attorney, Agent, or Firm* — Merchant & Gould P.C.

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(51) **Int. Cl.**
B65H 35/06 (2006.01)

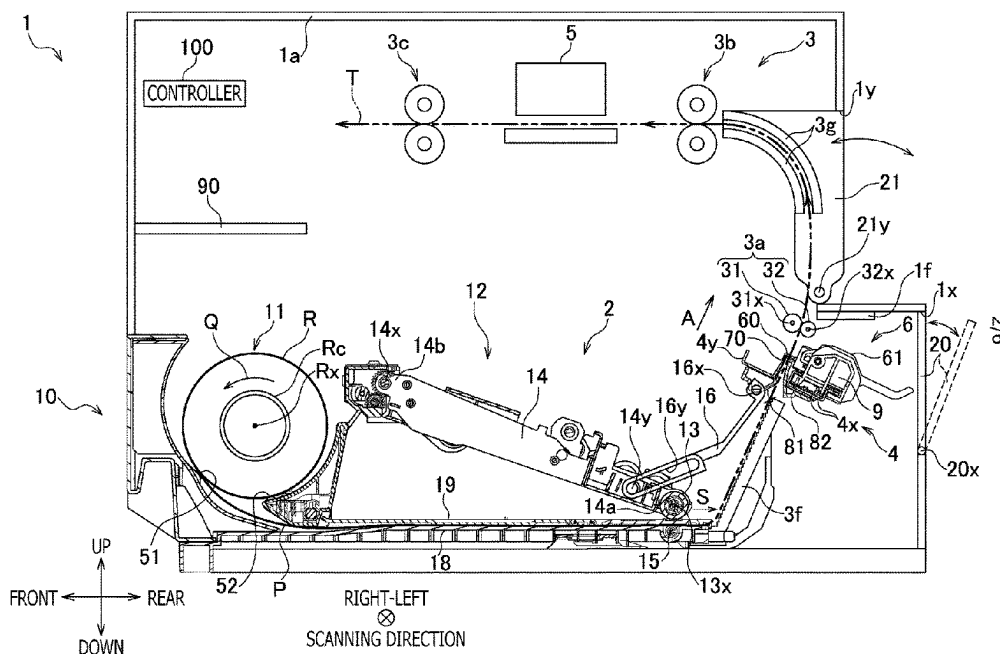
(52) **U.S. Cl.**
CPC **B65H 35/06** (2013.01); **B65H 2402/441**
(2013.01); **B65H 2407/10** (2013.01); **B65H**
2601/11 (2013.01)

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B65H 2407/10; B65H 2402/441; B65H
2801/11; B65H 2801/12

(57) **ABSTRACT**

An image recording apparatus includes a conveyer configured to convey a sheet-like medium along a conveying path in a conveying direction, a recorder configured to record an image on the sheet-like medium conveyed by the conveyer, a cutter unit including one or more cutters and configured to cut the sheet-like medium in the conveying path, and a scanning mechanism configured to move at least one of the one or more cutters included in the cutter unit in a scanning direction parallel to the sheet-like medium lying along the conveying path and intersecting with the conveying direction. The scanning mechanism includes a support member extending in the scanning direction and configured to support the cutter unit. The conveyer includes a path member fixed to the support member and defining a portion of the conveying path upstream of the cutter unit in the conveying direction.

14 Claims, 6 Drawing Sheets



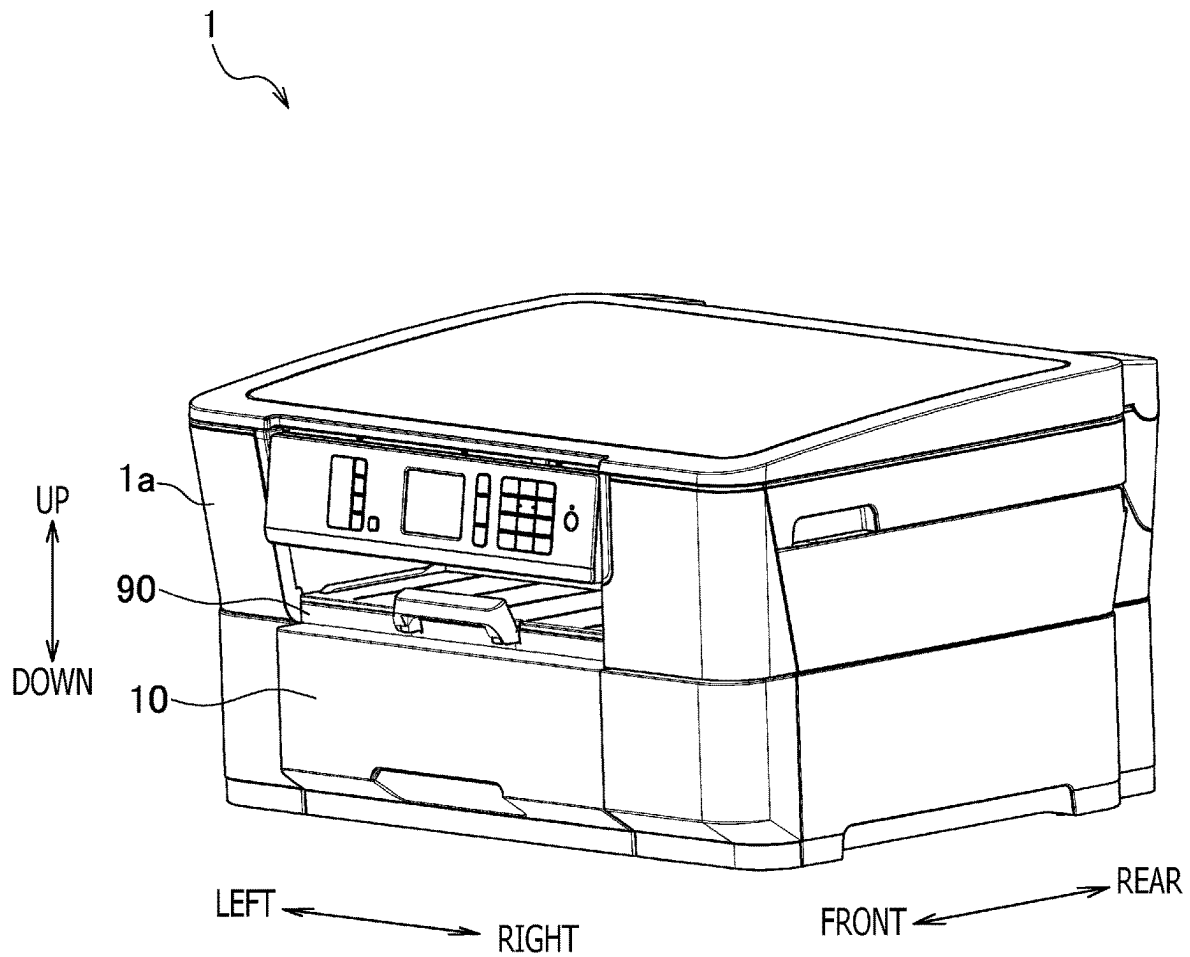
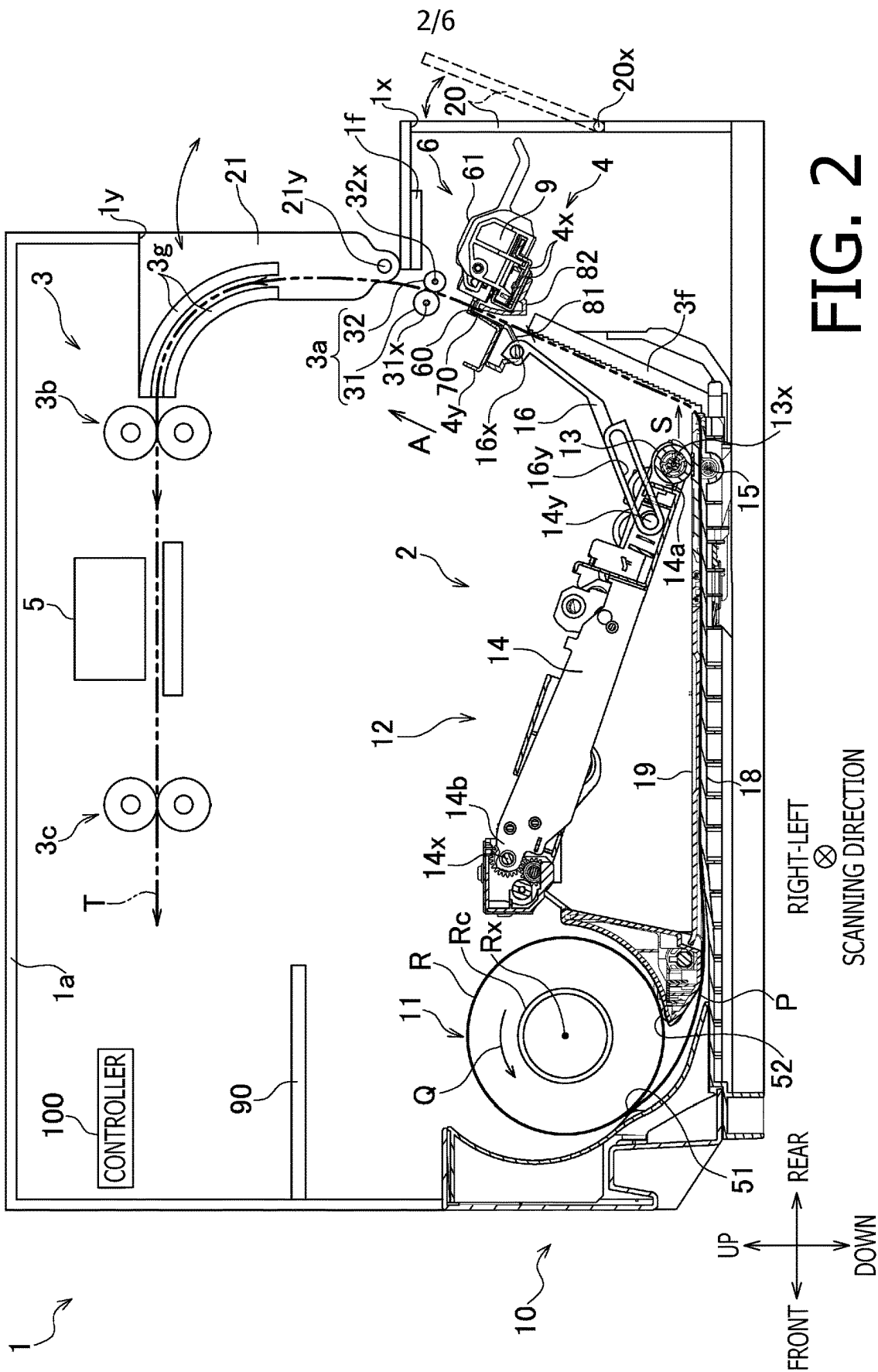


FIG. 1



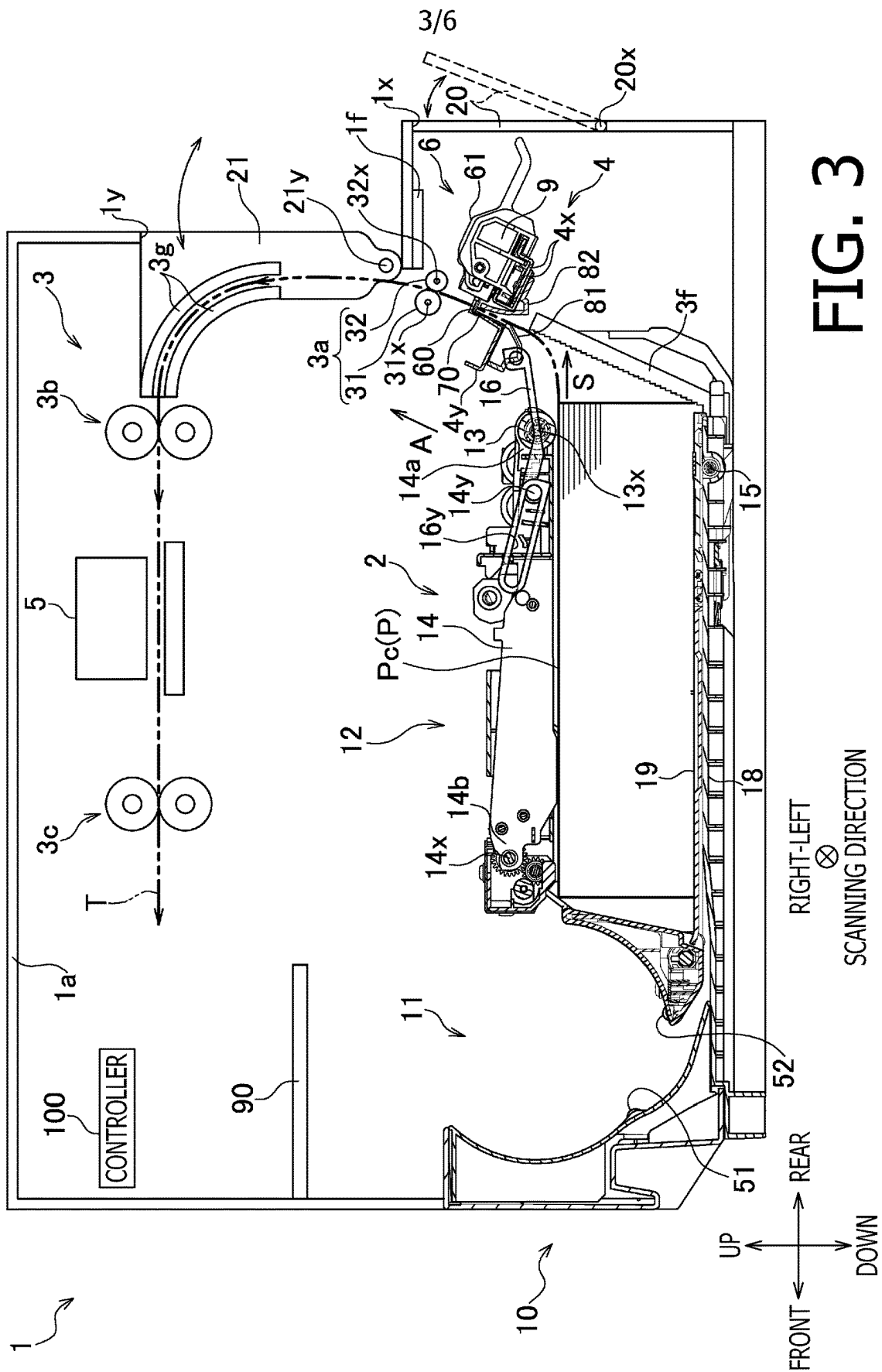
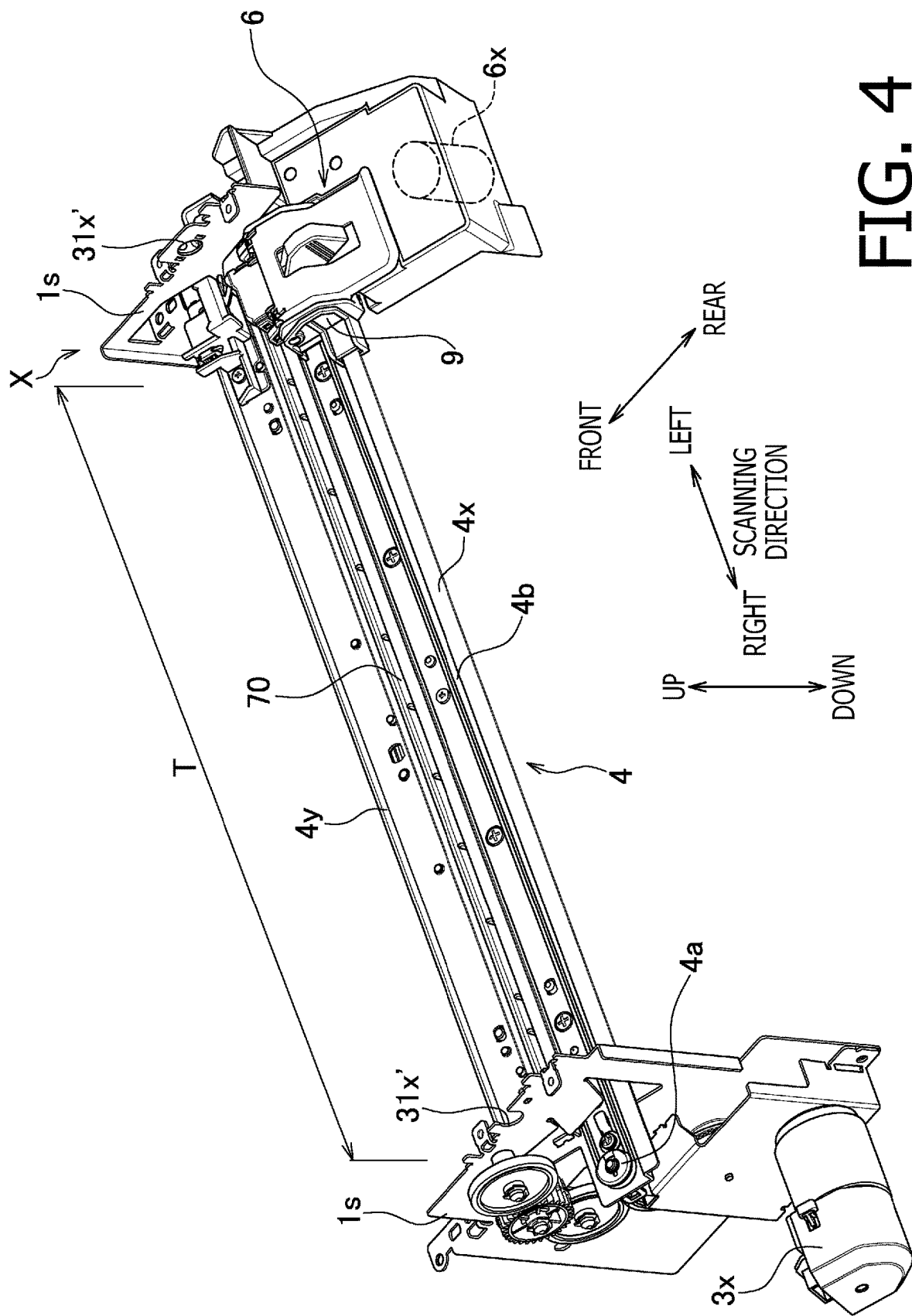
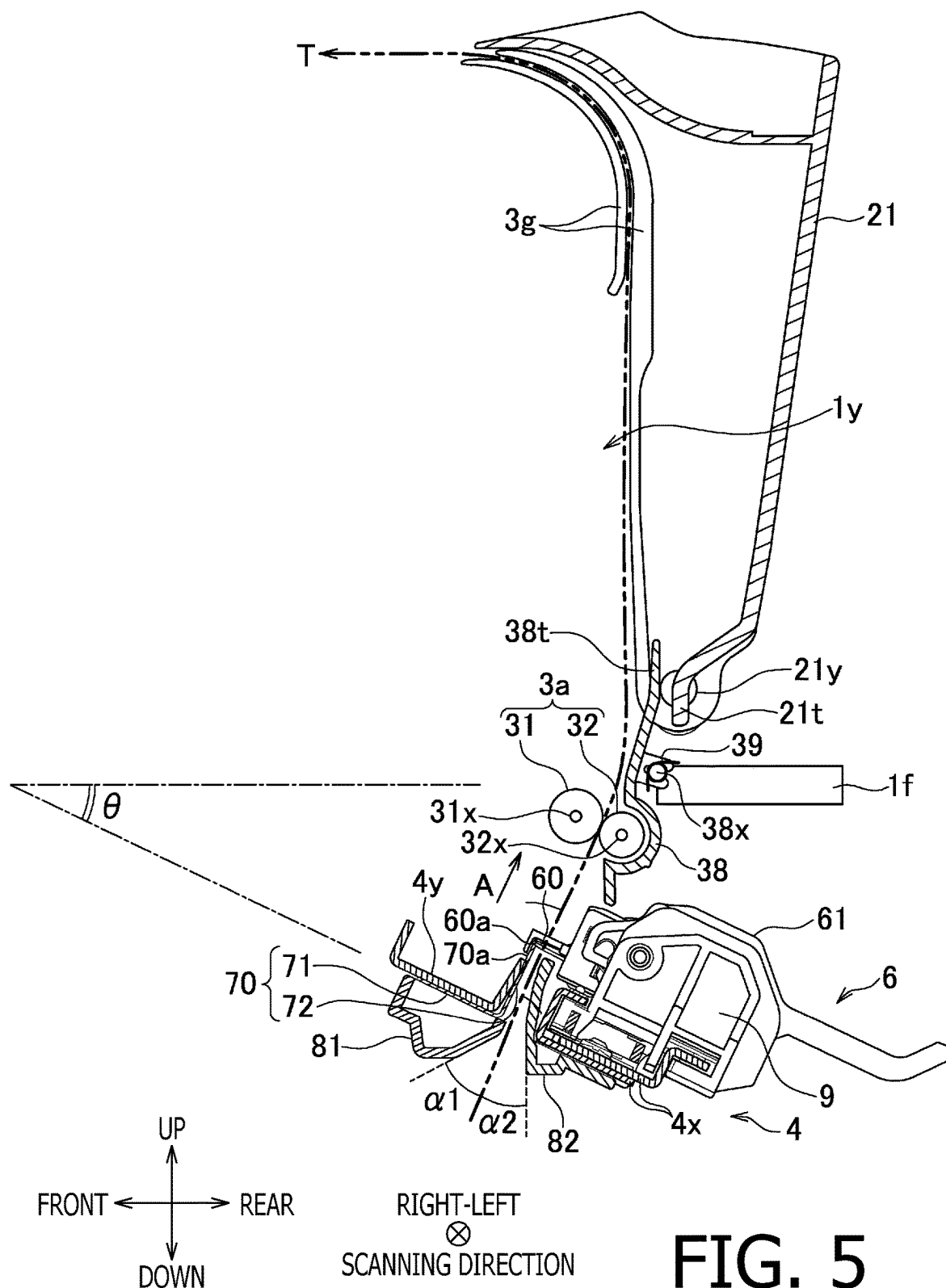
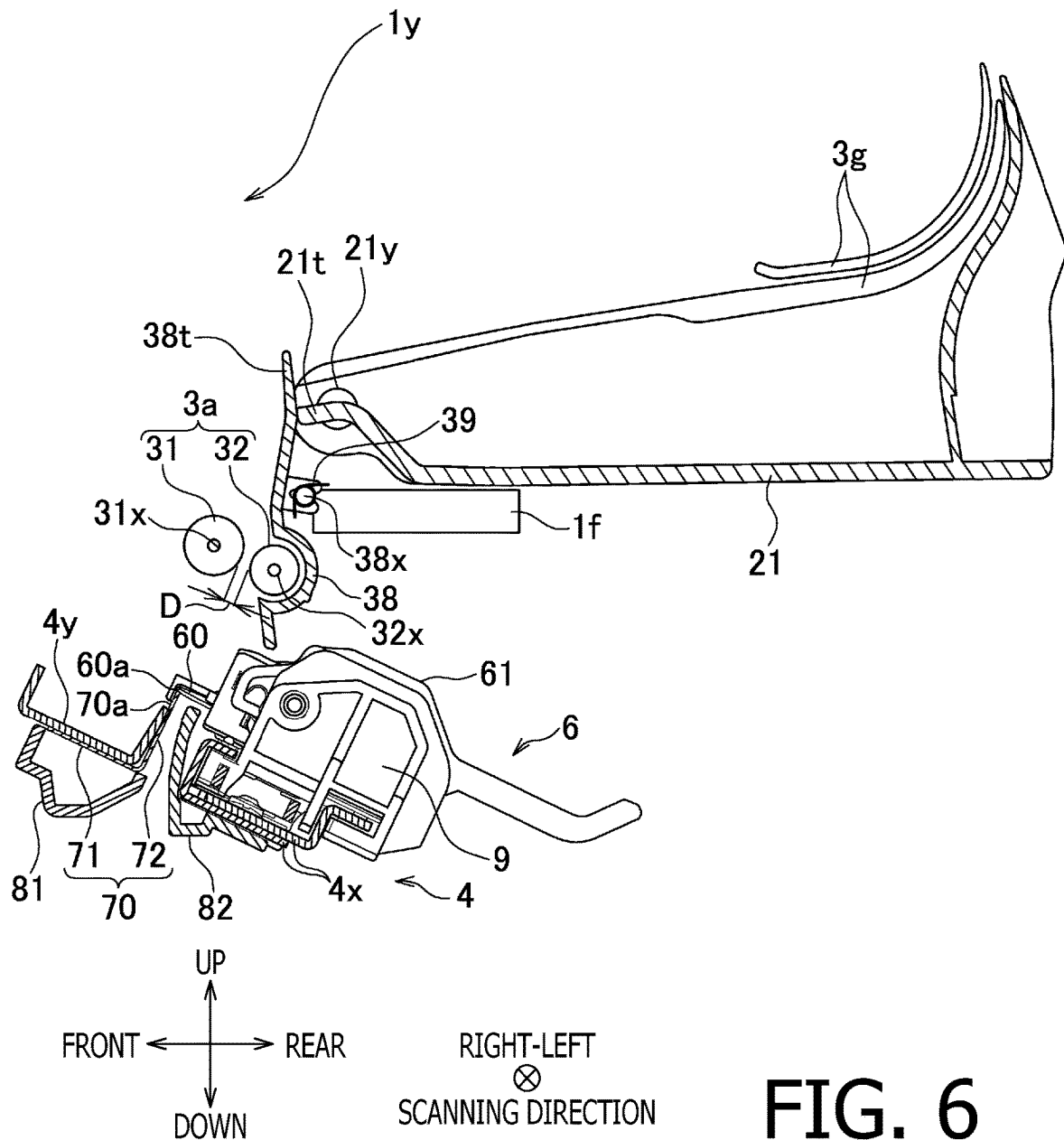


FIG. 3







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IMAGE RECORDING APPARATUS**REFERENCE TO RELATED APPLICATIONS**

This application claims priority from Japanese Patent Application No. 2022-037976 filed on Mar. 11, 2022 and Japanese Patent Application No. 2023-016784 filed on Feb. 7, 2023. The entire contents of the priority applications are incorporated herein by reference.

BACKGROUND ART

Aspects of the present disclosure relate to an image recording apparatus including a cutter configured to cut a sheet-like medium.

There is known an image recording apparatus configured to record an image while conveying a long sheet-like medium such as sheet, cloth, or labels along a conveying path. Such image recording apparatus may include a cutter configured to cut the sheet-like medium. For example, there is known an image recording apparatus having a configuration in which a sheet cutting device including a cutter is disposed downstream of an image forming unit (recorder) in the conveying direction.

DESCRIPTION

If the sheet-like medium jams near the cutter, a user may injure his/her fingers with the cutter when dealing with jamming. Therefore, careful operation is required and time and effort to deal with the jam increases. Such a problem is not considered in conventional image recording apparatuses.

At least one aspect of the present disclosure is advantageous to provide an image recording apparatus that makes it possible to suppress time and effort to deal with the jam.

According to aspects of the present disclosure, there is provided an image recording apparatus including a conveyor configured to convey a sheet-like medium along a conveying path in a conveying direction, a recorder configured to record an image on the sheet-like medium conveyed by the conveyor, a cutter unit including one or more cutters and configured to cut the sheet-like medium in the conveying path, and a scanning mechanism configured to move at least one of the one or more cutters included in the cutter unit in a scanning direction parallel to the sheet-like medium lying along the conveying path and intersecting with the conveying direction. The scanning mechanism includes a support member extending in the scanning direction and configured to support the cutter unit. The conveyor includes a path member fixed to the support member and defining a portion of the conveying path upstream of the cutter unit in the conveying direction.

According to aspects of the present disclosure, there is further provided an image recording apparatus including a tray configured to accommodate a sheet-like medium, a conveyor configured to convey the sheet-like medium accommodated in the tray along a conveying path in a conveying direction, a recorder configured to record an image on the sheet-like medium conveyed by the conveyor, and a cutter unit including one or more cutters and configured to cut the sheet-like medium in the conveying path. The conveyor includes a feeder configured to feed the sheet-like medium accommodated in the tray in a feeding direction toward the conveying path, and a pair of path members opposing to each other across the conveying path. The pair of path members define a portion of the conveying path between the tray and the cutter unit and include a first path

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member disposed upstream of the portion of the conveying path between the tray and the cutter unit in the feeding direction, and a second path member disposed downstream of the portion of the conveying path between the tray and the cutter unit in the feeding direction. An angle on an acute side between the first path member and the portion of the conveying path between the tray and the cutter unit is larger than an angle on an acute side between the second path member and the portion of the conveying path between the tray and the cutter unit.

According to aspects of the present disclosure, there is further provided an image recording apparatus including a conveyor configured to convey a sheet-like medium along a conveying path in the conveying direction, a recorder configured to record an image on the sheet-like medium conveyed by the conveyor, a cutter unit including one or more cutters and configured to cut the sheet-like medium in the conveying path, a housing in which the conveying path is formed, an opening formed to the housing, and a cover configured to move relative to the housing between a closed position for closing the opening and an open position for opening the opening. The conveyor includes an intermediate roller disposed either upstream or downstream of the cutter unit in the conveying direction and including two rotating bodies and configured to move between a contact position where the two rotating bodies are in contact with each other and a separated position at which the two rotating bodies are spaced apart from each other. The opening exposes the intermediate roller to the outside of the housing. When the cover moves from the closed position to the open position, the intermediate roller moves from the contact position to the separated position.

FIG. 1 is a perspective view of a printer.

FIG. 2 is a schematic side view showing an internal structure of the printer.

FIG. 3 is a schematic side view corresponding to FIG. 2 and showing a state in which cut sheets are accommodated in a sheet feed tray of the printer.

FIG. 4 is a perspective view of a cutter unit and a scanning mechanism included in the printer.

FIG. 5 is a side cross-sectional view showing a state in which a cover is at a closed position and a roller pair is at a contact position.

FIG. 6 is a side cross-sectional view showing a state in which the cover is at an open position and the roller pair is at a separated position.

OVERALL PRINTER CONFIGURATION

As shown in FIG. 1, a printer 1 (image recording apparatus) according to an embodiment of the present disclosure includes a housing 1a, a sheet feed tray 10 removable from the housing 1a, and a sheet discharge tray 90.

As shown in FIG. 2, the printer 1 further includes a conveyor 3, a head 5, a cutter unit 6, a scanning mechanism 4, and a controller 100. Elements of the conveyor 3 other than rollers 51 and 52 described later, the head 5, the cutter unit 6, the scanning mechanism 4, and the controller 100 are supported by the housing 1a. The rollers 51 and 52 are supported by the sheet feed tray 10.

The conveyor 3 is configured to convey the sheet P along a conveying path T in a conveying direction. The conveyor 3 includes rollers 51 and 52, a roller 13, an arm 14, a guide member 16, a pair of path members 81 and 82, roller pairs 3a-3c, a separating member 3f, a pair of guide members 3g, and a conveying motor 3x (see FIG. 8) configured to drive the rollers.

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The rollers **51** and **52**, the roller **13**, and the arm **14** of the conveyer **3** constitute a feeder **2** configured to feed the sheet **P** accommodated in the sheet feed tray **10** toward the conveying path **T**.

The conveying path **T** is a path that extends from the sheet feed tray **10** to the sheet discharge tray **90** while passing between the pair of guide members **3g** and under the head **5**. The conveying path **T** includes a U-shaped part at a portion upstream of the head **5** in the conveying direction. The separating member **3f**, the cutter unit **6**, the roller pair **3a**, the scanning mechanism **4**, and the pair of guide members **3g** are disposed in the U-shaped part. The cutter unit **6**, the scanning mechanism **4** and the roller pair **3a** are disposed at a bottom portion of the U-shaped part. The cutter unit **6** and the scanning mechanism **4** are disposed downstream of the separating member **3f** in the conveying direction and upstream of the roller pair **3a** in the conveying direction.

The sheet feed tray **10** includes a roll sheet accommodating part **11** configured to accommodate a roll sheet **R** (see FIG. 2) and a cut sheet accommodating part **12** configured to accommodate a plurality of cut sheets **Pc** (see FIG. 3) stacked in an up-down direction. The cut sheet **Pc** accommodated in the cut sheet accommodating part **12** is supported by an upper surface of a support plate **19** disposed along a bottom plate **18** of the sheet feed tray **10**.

The cut sheets **Pc** are removed from the cut sheet accommodating part **12** when using the roll sheet **R** (see FIG. 2), and the roll sheet **R** is removed from the roll sheet accommodating part **11** when using the cut sheets **Pc** (see FIG. 3). The cut sheet accommodating part **12** is a space behind the roll sheet accommodating part **11**.

In the present disclosure, the sheet **P** is used as a general term for sheet unwound from the roll sheet **R** and the cut sheet **Pc**. The cut sheet **Pc** is a sheet having a length along the conveying path **T** shorter than that of the sheet constituting the roll sheet **R**.

The roll sheet **R** is formed by winding a long sheet **P** in a roll shape on an outer peripheral surface of a cylindrical core member **Rc**. The roll sheet **R** is accommodated in the roll sheet accommodating part **11** with its rotation axis **Rx** (a center axis of the core member **Rc**) extending in the right-left direction.

The rollers **51** and **52** are disposed at the bottom of the roll sheet accommodating part **11**. The rollers **51** and **52** are rotatable about respective axes extending in the right-left direction. When the roll sheet **R** is accommodated in the roll sheet accommodating part **11**, an outer peripheral surface of a lower portion of the roll sheet **R** is supported by the rollers **51** and **52**. When setting the roll sheet **R**, the roll sheet **R** is manually rotated in a direction indicated by an arrow **Q** in FIG. 2 to unwind the sheet **P** from the roll sheet **R**. Then, the sheet **P** is made to pass through a gap between a lower surface of the support plate **19** and an upper surface of the bottom plate **18**, and a leading edge of the sheet **P** is nipped between the roller **13** and the roller **15**. In this state, when the conveying motor **3x** (see FIG. 4) is driven under the control of the controller **100** and the rollers **51**, **52** and **13** rotate, the sheet **P** unwound from the roll sheet **R** is fed backward (in a feeding direction **S**).

The roller **13** is supported by one end **14a** of the arm **14** and is rotatable about a shaft **13x** extending in the right-left direction. The other end **14b** of the arm **14** is supported by the housing **1a** via a shaft **14x** extending in the right-left direction. The arm **14** is swingable about the shaft **14x** with the other end **14b** as a fulcrum (see FIGS. 2 and 3).

In a state where the sheet feed tray **10** is mounted to the housing **1a**, when no cut **Pc** sheet is accommodated in the cut

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sheet accommodating part **12**, the roller **13** comes into contact with the upper surface of the support plate **19** (see FIG. 2). In a state where the sheet feed tray **10** is mounted to the housing **1a**, when one or more cut sheets **Pc** are accommodated in the cut sheet accommodating part **12**, the roller **13** comes into contact with the uppermost cut sheet **Pc** among the cut sheets **Pc** accommodated in the cut sheet accommodating part **12** (see FIG. 3). In this state, the cut sheet **Pc** is fed backward (i.e., in a feeding direction **S**) as the conveying motor **3x** (see FIG. 4) is driven under the control of the controller **100** and the roller **13** rotates.

The sheet **P** fed from the sheet feed tray **10** by the roller **13** (i.e., the sheet **P** unwound from the roll sheet **R** accommodated in the roll sheet accommodating part **11** or the cut sheet **Pc** accommodated in the cut sheet accommodating part **12**) comes into contact with the separating member **3f**, moves along the separating member **3f**, and is guided toward the roller pair **3a**.

The separating member **3f** is a member configuring a downstream end of the sheet feed tray **10** in the feeding direction **S** and defines a portion of the conveying path **T**.

The guide member **16** has a bent shape and has one end provided with a shaft **16x** extending in the right-left direction and the other end provided with an elongated hole **16y**. The shaft **16x** is supported by the path member **81**. A projection **14y** formed near one end **14a** of the arm **14** is inserted into the elongated hole **16y**. As shown in FIGS. 2 and 3, the guide member **16** swings about the shaft **16x** as the arm **14** swings and the projection **14y** moves along the elongated hole **16y**. The shaft **16x** is provided between the sheet feed tray **10** and the cutter unit **6** in the conveying path **T**. The guide member **16** defines the conveying path **T** upstream of the pair of path members **81** and **82** in the conveying direction.

The separating member **3f** is disposed behind the roller **13** and extends in an oblique direction intersecting with both the up-down direction and the front-rear direction. On a surface of the separating member **3f**, fine irregularities repeating along the conveying path **T** are formed. The irregularities prevent double conveyance (i.e., a phenomenon in which a plurality of cut sheets **Pc** are conveyed in an overlapped state). In other words, the separating member **3f** has a function of separating the cut sheet **Pc** in contact with the roller **13** from the other cut sheets **Pc**.

The head **5** includes a plurality of conventionally-known nozzles formed on a lower surface of the head **5**, and a conventionally-known driver **IC**. When the sheet **P** conveyed by the conveyer **3** passes through a position facing the lower surface of the head **5**, the driver **IC** is driven under the control of the controller **100**, so that ink is ejected from the nozzles and lands on the sheet **P**, and an image is recorded on the sheet **P**. The head **5** may be of either a line type that ejects ink from the nozzles while the position is fixed or a serial type that ejects ink from the nozzles while moving in the right-left direction.

The cutter unit **6** is configured to cut the sheet **P** unwound from the roll sheet **R** between the separating member **3f** and the roller pair **3a** in the conveying path **T**. The cutter unit **6** includes cutter **60** and **70**, and a holder **61** configured to hold the cutter **60**. The holder **61** is detachably attached to the carriage **9**.

The carriage **9** can be made to reciprocate in a scanning direction (right-left direction) by the scanning mechanism **4**. The scanning direction is parallel to the sheet **P** lying along the conveying path **T** and intersects with a conveying direction **A** (See FIGS. 2 and 3. A conveying direction in the vicinity of the cutter unit **6**).

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As shown in FIG. 4, the scanning mechanism 4 includes a pair of guide rails 4x and 4y extending in the scanning direction, a pair of pulleys 4a (only one of which is shown in FIG. 4) disposed to be spaced apart from each other in the scanning direction, a belt 4b wound around the pair of pulleys 4a and having the carriage 9 fixed thereto, and a cutting motor 6x. The guide rail 4y is located in front of the guide rail 4x and is spaced apart from the guide rail 4x with a small gap. The conveying path T (see FIGS. 2 and 3) is formed in this gap.

As shown in FIG. 4, the guide rails 4x and 4y are fixed to a pair of side frames 1s with screws or the like. The pair of side frames 1s are portions of the housing 1a and are made of metal or the like. The pair of side frames 1s are spaced apart from each other in the right-left direction and support the left and right ends of the guide rails 4x and 4y.

The pair of pulleys 4a, the belt 4b, and the cutting motor 6x are attached to the guide rail 4x. The guide rail 4x supports the carriage 9 fixed to the belt 4b, and the holder 61 and the cutter 60 mounted to the carriage 9.

The guide rail 4y supports the cutter 70. As shown in FIG. 5, the cutter 70 is L-shaped and has a portion 71 that covers a lower surface of the guide rail 4y and a portion 72 that covers a side surface of the guide rail 4y that defines the conveying path T. The portion 72 of the cutter 70 extends along the conveying path T. The cutter 70 has a blade 70a at an end portion of the portion 72 on the downstream side in the conveying direction A.

While the cutter 70 is fixed, the cutter 60 rotates. The cutter 60 is disk-shaped and has a blade 60a at its peripheral edge. When cutting the sheet P, the blade 60a of the cutter 60 extends in a direction intersecting with the sheet P lying along the conveying path T downstream of the blade 70a of the cutter 70 in the conveying direction A, and overlaps with the blade 70a of the cutter 70 in the conveying direction A.

When the cutting motor 6x is driven under the control of the controller 100, the belt 4b travels in the scanning direction, and the carriage 9 and the holder 61 move from an attaching/detaching position X outside the conveying path T into the conveying path T. At this time, the cutter 60 rotates by the driving of the cutting motor 6x. The sheet P unwound from the roll sheet R is cut in a width direction of the sheet P by the fixed cutter 70 and the cutter 60 rotating while moving in the scanning direction.

The attaching/detaching position X is at left ends of the guide rails 4x and 4y, which is a home position of the movement of the carriage 9 as described above, and is also a position for attaching/detaching the holder 61 to/from the carriage 9.

As shown in FIGS. 2 and 3, the pair of path members 81 and 82 are disposed so as to oppose each other across the conveying path T, and define a portion of the conveying path T upstream of the cutter unit 6 in the conveying direction (i.e., a portion between the sheet feed tray 10 and the cutter unit 6). The path member 81 is located in front of, upstream in the feeding direction S (see FIG. 3) of, and downstream in a winding direction of the sheet P constituting the roll sheet R (a direction shown by an arrow Q shown in FIG. 2) of, a portion of the conveying path T defined by the pair of path members 81 and 82 (i.e., a portion between the pair of path members 81 and 82). The path member 82 is located behind, and downstream in the feeding direction S (see FIG. 3) of, the portion of the conveying path T defined by the pair of path members 81 and 82.

The path member 81 is fixed to the guide rail 4y, and the path member 82 is fixed to the guide rail 4x.

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Furthermore, as shown in FIG. 5, an angle $\alpha 1$ (an angle on the acute side) between the path member 81 and the portion of the conveying path T defined by the pair of path members 81 and 82 (i.e., a portion between the pair of path members 81 and 82) is larger than an angle $\alpha 2$ (an angle on the acute side) between the path member 82 and the portion of the conveying path T defined by the pair of path members 81 and 82.

As shown in FIG. 5, the portion of the conveying path T defined by the pair of path members 81 and 82 is orthogonal to the cutter 60. Furthermore, as shown in FIGS. 2 and 3, the portion of the conveying path T defined by the pair of path members 81 and 82 is extending along the conveying direction A, is parallel to a portion of the conveying path T defined by the separating member 3f, and is parallel to a portion of the conveying path T defined by later-described two rotating bodies 31 and 32. The angles $\alpha 1$ and $\alpha 2$ are angles between the surfaces of the path members 81 and 82 facing the conveying path T and the portion of the conveying path T defined by the pair of path members 81 and 82.

In view of preventing jamming, the path members 81 and 82 are designed such that an angle on the acute side between the surface of the path member 81 facing the conveying path T and a leading edge of the sheet P unwound from the roll sheet R and abutting the path member 81 (a first entry angle) and an angle on the acute side between the surface of the path member 82 facing the conveying path T and a leading edge of the cut sheet Pc abutting the path member 82 (a second entry angle) become equal to or less than a prescribed angle (e.g., 63 degrees). In the present embodiment, the first entry angle is about 50 degrees, and the second entry angle is about 55 to 58 degrees. That is, the first entry angle is smaller than the second entry angle.

As shown in FIGS. 2 and 3, the roller pair 3a is disposed downstream of the cutter unit 6 in the conveying direction, and includes two rotating bodies 31 and 32. The two rotating bodies 31 and 32 are rotatable about rotation shafts 31x and 32x extending in the right-left direction, respectively.

The rotation shaft 31x of the rotating body 31 disposed in front of the rotating body 32 is supported by a pair of side frames 1s (see FIG. 4). As shown in FIG. 4, the right and left ends of the rotation shaft 31x are inserted into holes 31x' formed to the pair of side frames 1s.

As shown in FIGS. 5 and 6, the rotating body 32 disposed behind the rotating body 31 is rotatably supported by a holder 38. The holder 38 is swingable about a swing axis 38x in the right-left direction. As the holder 38 swings, the rotating body 32 supported by the holder 38 swings. Thus, the roller pair 3a can move between a contact position where the two rotating bodies 31 and 32 are in contact with each other (see FIG. 5) and a separated position where the two rotating bodies 31 and 32 are spaced apart from each other (see FIG. 6). A distance D between the two rotating bodies 31 and 32 when the roller pair 3a is at the separated position is such that a human finger does not enter (e.g., 5 mm or less).

A torque spring 39 is attached to a shaft 38x of the holder 38. The torque spring 39 urges the holder 38 in a direction in which the holder 38 swings clockwise in FIGS. 5 and 6 about the shaft 38x (i.e., in a direction in which the roller pair 3a moves from the separated position to the contact position).

The shaft 38x of the holder 38 is located downstream of a rotation shaft 32x of the rotating body 32 in the conveying direction A, and is supported by a frame 1f.

The frame 1f is a part of the housing 1a and is made of metal or the like. The frame 1f extends in a horizontal

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direction (i.e., a direction orthogonal to the up-down direction). In contrast, the guide rails **4x** and **4y** extend in an oblique direction intersecting with both the up-down direction and the front-rear direction, and are inclined downward toward the rear. Thus, an angle θ between the frame **1f** and the guide rails **4x** and **4y** is an acute angle (see FIG. 5).

As shown in FIGS. 2 and 3, two openings **1x** and **1y** arranged in the up-down direction are formed to a rear wall of the housing **1a**. Through the lower opening **1x**, a portion inside the housing **1a** including the attaching/detaching position X (i.e., the carriage **9** at the attaching/detaching position X: see FIG. 4) is exposed to the outside of the housing **1a**. Through the upper opening **1y**, the roller pair **3a** is exposed to the outside of the housing **1a**. The openings **1x** and **1y** are formed at portions of the housing **1a** that face the bottom portion of the U-shaped part of the conveying path T.

As shown in FIGS. 2 and 3, a cover **20** configured to cover the opening **1x** and a cover **21** configured to cover the opening **1y** are attached to the rear wall of the housing **1a** movably relative to the housing **1a**.

The cover **20** is swingable about a shaft **20x** provided at a lower end of the opening **1x** and extending along the right-left direction, and can move between a closed position for closing the opening **1x** (see the solid line in FIG. 2) and an open position for opening the opening **1x** (see the broken line in FIG. 2) by swinging about the shaft **20x**. By placing the cover **20** at the open position (see the broken line in FIG. 2), a space between the frame **1f** and the guide rails **4x** and **4y** can be accessed through the opening **1x**. Therefore, it becomes possible to attach and detach the holder **61** to and from the carriage **9** at the attaching/detaching position X, and it becomes possible to deal with jamming.

The cover **21** is swingable about a shaft **21y** provided at a lower end of the opening **1y** and extending along the right-left direction, and can move between a closed position for closing the opening **1y** (see FIG. 5) and an open position for opening the opening **1y** (see FIG. 6) by swinging about the shaft **21y**.

When the cover **21** is at the closed position (see FIG. 5), an upper end **38t** of the holder **38** and a lower end **21t** of the cover **21** overlap with each other in the front-rear direction and are spaced apart from each other in the front-rear direction with a slight gap therebetween. When the cover **21** moves from the closed position (see FIG. 5) to the open position (see FIG. 6), the lower end **21t** of the cover **21** comes into contact with the upper end **38t** of the holder **38** to move the upper end **38t** forward against the urging force of the torque spring **39**. At this time, the roller pair **3a** moves from the contact position (see FIG. 5) to the separated position (see FIG. 6) as the holder **38** swings about the shaft **38x** and the rotating body **32** held by the holder **38** moves backward.

When the cover **21** moves from the open position (see FIG. 6) to the closed position (see FIG. 5), the lower end **21t** of the cover **21** is separated from the upper end **38t** of the holder **38**, and the holder **38** swings clockwise in FIGS. 5 and 6 about the shaft **38x** by the urging force of the torque spring **39**. As a result, the roller pair **3a** moves from the separated position (see FIG. 6) to the contact position (see FIG. 5).

As described above, the roller pair **3a** moves between the contact position and the separated position in conjunction with the opening/closing motion of the cover **21**.

The cover **21** is provided with a pair of guide members **3g**. When the cover **21** is at the closed position (see FIG. 5), the cover **21** forms the conveying path T. By placing the cover

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21 at the open position (see FIG. 6) and placing the roller pair **3a** at the separated position, the sheet P jammed around the cutters **60** and **70** can be taken out of the housing **1a** through the opening **1y** through the gap between the two rotating bodies **31** and **32** with the distance D and between the pair of guide members **3g**.

As described above, according to the present embodiment, the path members **81** and **82** are fixed to the guide rails **4x** and **4y** to position the path members **81** and **82** with respect to the guide rails **4x** and **4y** (see FIGS. 2 and 3). Therefore, the sheet P can be smoothly conveyed from the path members **81** and **82** toward the cutter unit **6** supported by the guide rails **4x** and **4y**, and thus jamming is less likely to occur. Accordingly, it is possible to reduce the time and effort to deal with the jam.

According to the present embodiment, the angle $\alpha 1$ between the path member **81** and the portion of the conveying path T defined by the pair of path members **81** and **82** is larger than the angle $\alpha 2$ between the path member **82** and the portion of the conveying path T defined by the pair of path members **81** and **82** (see FIG. 5). For example, when conveying the sheet P unwound from the roll sheet R with the conveyer **3** (see FIG. 2), the leading edge of the sheet P is likely to contact the path member **81**. In this case, since the angle $\alpha 1$ is large, a path width of an entry portion between the pair of path members **81** and **82** is large, and thus it is easier to guide the leading edge of the sheet P to between the pair of path members **81** and **82**. Therefore, the sheet P can be appropriately guided and thus jamming is less likely to occur. Accordingly, it is possible to suppress the time and effort to deal with the jam. Furthermore, for example, when feeding an uppermost cut sheet Pc among the cut sheets Pc accommodated in the cut sheet accommodating part **12** with the feeder **2** (see FIG. 3), if the angle $\alpha 1$ is small, a leading edge of the fed uppermost cut sheet Pc may come into contact with the path member **81** and jamming may occur. In this respect, according to the present embodiment, since the angle $\alpha 1$ is large, the contact of the leading edge of the uppermost cut sheet Pc with the path member **81** can be suppressed and thus jamming is less likely to occur. Accordingly, it is possible to suppress the time and effort to deal with the jam.

According to the present embodiment, when the cover **21** is moved from the closed position (see FIG. 5) to the open position (see FIG. 6), the roller pair **3a** moves from the contact position (see FIG. 5) to the separated position (see FIG. 6), and the sheet P jammed around the cutters **60** and **70** can be taken out of the housing **1a** through the gap between the two rotating bodies **31** and **32** with the distance D. When the cover **21** is moved from the open position (see FIG. 6) to the closed position (see FIG. 5) after dealing with jamming, the roller pair **3a** moves from the separated position (see FIG. 6) to the contact position (see FIG. 5), and the recording process can be executed. By linking the position of the roller pair **3a** with the opening and closing operation of the cover **21**, it is possible to suppress the time and effort to deal with the jam.

The path member **81** is located downstream of a portion of the conveying path T defined by the pair of path members **81** and **82** (i.e., the portion between the pair of path members **81** and **82**) in the winding direction of the sheet P constituting the roll sheet R (the direction indicated by the arrow Q shown in FIG. 2). Due to this configuration, a leading edge of the sheet P unwound from the roll sheet R easily comes into contact with the path member **81**, and thus the sheet P can be appropriately guided by the path member **81**. There-

fore, the sheet P can be smoothly conveyed from the path member **81** toward the cutter unit **6**, and thus jamming is less likely to occur.

The path member **82** is located downstream of the portion of the conveying path T defined by the pair of path members **81** and **82** (i.e., the portion between the pair of path members **81** and **82**) in the feeding direction S (see FIG. 3). Due to this configuration, the leading edge of the cut sheet Pc fed from the cut sheet accommodating part **12** can easily come into contact with the path member **82**, and thus the cut sheet Pc can be appropriately guided by the path member **82**. Therefore, the sheet P can be smoothly conveyed from the path member **82** toward the cutter unit **6**, and thus jamming is less likely to occur.

The printer **1** includes the pair of path members **81** and **82** opposing to each other across the conveying path T. Due to this configuration, the sheet P can be smoothly conveyed toward the cutter unit **6** by the pair of path members **81** and **82**, and thus jamming is less likely to occur.

The shaft **16x** of the guide member **16** is fixed to the path member **81**, so that the guide member **16** is positioned with respect to the path member **81** (see FIGS. 2 and 3). Due to this configuration, the sheet P can be smoothly conveyed from the guide member **16** toward the path member **81**, and thus jamming is less likely to occur.

The guide rails **4x** and **4y** and the rotation shaft **31x** of the rotating body **31** are fixed to the same member (i.e., the side frame **1s** shown in FIG. 3), and thus the guide rails **4x** and **4y** are positioned with respect to the rotating body **31**. Due to this configuration, the conveyance of the sheet P between the cutter unit **6** supported by the guide rails **4x** and **4y** and the roller pair **3a** becomes smooth, and thus jamming is less likely to occur.

The distance D (see FIG. 6) between the two rotating bodies **31** and **32** when the roller pair **3a** is at the separated position is such that a human finger does not enter. Due to this configuration, when taking the sheet P jammed around the cutters **60** and **70** out of the housing **1a** through the opening **1y**, the user's fingers will not reach the cutters **60** and **70** through the gap between the two rotating bodies **31** and **32**. Therefore, it is possible to prevent the user from injuring his or her finger with the cutters **60** and **70**.

The roller pair **3a** is disposed downstream of the cutter unit **6** in the conveying direction (see FIGS. 2 and 3). Due to this configuration, when the sheet P is jammed around the cutters **60** and **70**, the cutter unit **6** can cut the sheet P, and a portion of the sheet P downstream of a cutting position by the cutter unit **6** in the conveying direction can be taken out of the housing **1a** through the gap between the two rotating bodies **31** and **32**.

The printer **1** includes the holder **38** configured to rotatably support one (the rotating body **32**) of the two rotating bodies **31** and **32** constituting the roller pair **3a**, and the roller pair **3a** moves from the contact position (see FIG. 5) to the separated position (see FIG. 6) as the holder **38** comes into contact with the cover **21** and swings. Therefore, the position of the roller pair **3a** is more effectively linked with the opening and closing operation of the cover **21**.

The shaft **38x** of the holder **38** is located downstream of the rotation shaft **32x** of the rotating body **32** in the conveying direction A (see FIGS. 5 and 6). Due to this configuration, compared to a case where the shaft **38x** of the holder **38** is located upstream of the rotation shaft **32x** of the rotating body **32** in the conveying direction A, a space downstream of the cutter unit **6** in the conveying direction A can be secured, and thus it is easier to deal with jamming.

The frame **1f** and the guide rails **4x** and **4y** are arranged to form the acute angle θ (see FIG. 5). Due to this configuration, it is easy to access a space between the frame **1f** and the guide rails **4x** and **4y** to deal with jamming.

5 Modification

While the invention has been described in conjunction with various example structures outlined above and illustrated in the figures, various alternatives, modifications, variations, improvements, and/or substantial equivalents, whether known or that may be presently unforeseen, may become apparent to those having at least ordinary skill in the art. Accordingly, the example embodiments of the disclosure, as set forth above, are intended to be illustrative of the invention, and not limiting the invention. Various changes may be made without departing from the spirit and scope of the disclosure. Therefore, the disclosure is intended to embrace all known or later developed alternatives, modifications, variations, improvements, and/or substantial equivalents. Some specific examples of potential alternatives, modifications, or variations in the described invention are provided below:

In the above-described embodiment, the portion of the conveying path T defined by the pair of path members **81** and **82** is orthogonal to the cutter **60**, is parallel to a portion of the conveying path T defined by the separating member **3f**, and is parallel to a portion of the conveying path T defined by the two rotating bodies **31** and **32**. However, the portion of the conveying path T defined by the pair of path members **81** and **82** is not limited to the one that satisfies all of these conditions but may be the one that satisfies at least one of these conditions.

In the above-described embodiment, the cutter unit **6** includes the disk-shaped rotating cutter **60** and the fixed cutter **70**. However, the cutter unit **6** may include a pair of disk-shaped rotating cutters. Furthermore, the number of cutters included in the cutter unit **6** is not limited to two, but may be one or three or more.

In the above-described embodiment, the scanning mechanism **4** moves one of the cutters (i.e., the cutter **60**) included in the cutter unit. However, for example, the scanning mechanism **4** may move all the cutters included in the cutter unit (e.g., a pair of disk-shaped cutters configured to rotate).

In the above-described embodiment, the path members **81** and **82** are formed as a pair. However, for example, only one of the path members **81** and **82** may be provided.

In the above-described embodiment, the roller pair **3a** is disposed downstream of the cutter unit in the conveying direction. However, the roller pair **3a** may be disposed upstream of the cutter unit in the conveying direction.

The sheet P is not limited to paper, but may be cloth or plastic film. In other words, the sheet P may be made of any material as long as it is sheet-like.

The head **5** may discharge liquid other than ink (e.g., a treatment liquid for aggregating or precipitating components in the ink). The head **5** is not limited to a liquid discharge system, and may be a laser system, a thermal transfer system, or the like.

Aspects of the present disclosure may be applied not only to a printer but can also to a facsimile machine, a copy machine, a multi-function machine, and the like.

The sheet feed tray **10** is an example of a "tray" according to aspects of the present disclosure. The sheet P is an example of a "sheet-like medium" according to aspects of the present disclosure. The separating member **3f** is an example of a "wall" according to aspects of the present disclosure. The shaft **16x** is an example of a "first shaft" according to aspects of the present disclosure. The head **5** is

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an example of a “recorder” according to aspects of the present disclosure. The pair of guide rails **4x** and **4y** are examples of a “supporting member” according to aspects of the present disclosure. The path member **81** is an example of a “first path member” according to aspects of the present disclosure. The path member **82** is an example of a “second path member” according to aspects of the present disclosure. The roller pair **3a** is an example of an “intermediate roller” according to aspects of the present disclosure. The shaft **38x** is an example of a “second shaft” according to aspects of the present disclosure. The opening **1y** is an example of an “opening” according to aspects of the present disclosure.

What is claimed is:

1. An image recording apparatus comprising:

a conveyer configured to convey a sheet-like medium along a conveying path in a conveying direction;

a recorder configured to record an image on the sheet-like medium conveyed by the conveyer;

a cutter unit including one or more cutters and configured to cut the sheet-like medium in the conveying path; and a scanning mechanism configured to move at least one of the one or more cutters included in the cutter unit in a scanning direction parallel to the sheet-like medium lying along the conveying path and intersecting with the conveying direction,

wherein the scanning mechanism includes a support member extending in the scanning direction and configured to support the cutter unit, and

wherein the conveyer includes a path member extending from and fixed to the support member and defining a portion of the conveying path upstream of the cutter unit in the conveying direction.

2. The image recording apparatus according to claim 1, comprising a tray configured to accommodate a roll body formed by winding the sheet-like medium in a roll shape,

wherein the conveyer conveys the sheet-like medium unwound from the roll body accommodated in the tray along the conveying path, and

wherein the path member is located downstream of the portion of the conveying path defined by the path member in a winding direction of the sheet-like medium constituting the roll body.

3. The image recording apparatus according to claim 1, comprising a tray configured to accommodate a plurality of sheet-like medium in a stacked state,

wherein the conveyer includes a feeder configured to feed an uppermost sheet-like medium among the plurality of sheet-like medium accommodated in the tray in a feeding direction toward the conveying path, and

wherein the path member is located downstream of the portion of the conveying path defined by the path member in the feeding direction.

4. The image recording apparatus according to claim 1, comprising a pair of the path members opposing to each other across the conveying path.

5. The image recording apparatus according to claim 1, comprising a tray configured to accommodate a plurality of sheet-like medium in a stacked state,

wherein the conveyer includes:

a roller configured to come into contact with an uppermost sheet-like medium among the plurality of sheet-like medium accommodated in the tray and configured to feed the sheet-like medium;

an arm having one end configured to support the roller and the other end and configured to swing about the other end as a fulcrum; and

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a guide member configured to move in conjunction with the swinging of the arm, the guide member being configured to swing about a first shaft provided between the tray and the cutter unit in the conveying path and configured to define a portion of the conveying path upstream of the path member in the conveying direction,

wherein the first shaft is supported by the path member.

6. The image recording apparatus according to claim 1, wherein the conveyer includes an intermediate roller disposed either upstream or downstream of the cutter unit in the conveying direction and configured to rotate about a rotation shaft, and

wherein the image recording apparatus further includes a member that supports both the support member and the rotation shaft.

7. An image recording apparatus comprising:

a tray configured to accommodate a sheet-like medium;

a conveyer configured to convey the sheet-like medium accommodated in the tray along a conveying path in a conveying direction;

a recorder configured to record an image on the sheet-like medium conveyed by the conveyer; and

a cutter unit including one or more cutters and configured to cut the sheet-like medium in the conveying path, wherein the conveyer includes:

a feeder configured to feed the sheet-like medium accommodated in the tray in a feeding direction toward the conveying path; and

a pair of path members opposing to each other across the conveying path,

wherein the pair of path members define a portion of the conveying path between the tray and the cutter unit and include a first path member disposed upstream of the portion of the conveying path between the tray and the cutter unit in the feeding direction, and a second path member disposed downstream of the portion of the conveying path between the tray and the cutter unit in the feeding direction,

wherein an angle on an acute side between the first path member and the portion of the conveying path between the tray and the cutter unit is larger than an angle on an acute side between the second path member and the portion of the conveying path between the tray and the cutter unit.

8. The image recording apparatus according to claim 7, wherein the tray is configured to accommodate a roll body formed by winding the sheet-like medium in a roll shape,

wherein the conveyer conveys the sheet-like medium unwound from the roll body accommodated in the tray along the conveying path, and

wherein the first path member is located downstream of the portion of the conveying path between the tray and the cutter unit in a winding direction of the sheet-like medium constituting the roll body.

9. The image recording apparatus according to claim 7, wherein the tray is configured to accommodate a plurality of sheet-like medium in a stacked state,

wherein the feeder is configured to feed an uppermost sheet-like medium among the plurality of sheet-like medium accommodated in the tray.

10. The image recording apparatus according to claim 7, wherein the portion of the conveying path between the tray and the cutter unit is orthogonal to at least one of the one or more cutters.

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11. The image recording apparatus according to claim 7,
wherein the conveyer includes a wall configuring a down-
stream end of the tray in the feeding direction and
defining a portion of the conveying path, and
wherein the portion of the conveying path between the 5
tray and the cutter unit is parallel to the portion of the
conveying path defined by the wall.

12. The image recording apparatus according to claim 7,
wherein the conveyer includes an intermediate roller
disposed downstream of the cutter unit in the convey- 10
ing direction and including two rotating bodies, and
wherein the portion of the conveying path between the
tray and the cutter unit is parallel to a portion of the
conveying path defined by the two rotating bodies.

13. The image recording apparatus according to claim 1, 15
wherein the path member is disposed upstream of the
recorder in the conveying direction.

14. The image recording apparatus according to claim 13,
wherein the cutter unit is disposed upstream of the recorder
in the conveying direction. 20

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